

Vol 3 — Nuclear & Atomic Physics from D_IV⁵

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23) — chapter content TO BUILD Lead: Elie (primary, candidate) + Lyra (theoretical support) + Grace (catalog backbone) + Keeper (audit)

Reader target

Graduate physics with nuclear and atomic specialization. Reads Vol 0 (Substrate Foundation) + Vol 1 (QFT) + Vol 2 (Particle Physics) as prerequisite; then Vol 3 derives nuclear structure + atomic spectroscopy from substrate.

Prerequisites

- Vol 0 Substrate Foundation (D_IV⁵ + five integers + operator zoo + Strong-Uniqueness)
- Vol 1 QFT from D_IV⁵ (Bergman Hilbert space + Casimir + Hamiltonian + gauge fields)
- Vol 2 Particle Physics (Standard Model particles + masses + couplings)
- Standard graduate nuclear + atomic physics background (Krane / Bransden-Joachain level)

BST integer focus

Nuclear & atomic physics is where substrate spin-orbit coupling $C_2/n_C = 6/5$ and orbital sequence $(2l+1) = 1, N_c, n_C, g = 1, 3, 5, 7$ structure observed phenomenology directly. Vol 3 traces magic numbers, SEMF coefficients, hyperfine, Lamb shift, and atomic shell structure to specific BST integers.

Key integers active in Vol 3: - $C_2 = 6, n_C = 5$: spin-orbit coupling $\kappa_{ls} = C_2/n_C = 6/5 \rightarrow$ all 7 nuclear magic numbers - $N_c = 3, n_C = 5$: orbital degeneracy sequence $(2l+1) = 1, 3, 5, 7$ - $g = 7$: SEMF surface coefficient $a_S = (g+1) \cdot B_d = 8\alpha m_p / \pi$ - $N_{max} = 137$: atomic ceiling ($Z \leq N_{max}$ related to $\alpha = 1/N_{max}$)

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Existing BST coverage	Key theorems / toys
1	Nuclear Substrate Reading — D_IV ⁵ scale → nuclear physics emergent	~70% (substrate-as-nuclear-arena)	T2418 $\Lambda \leftrightarrow$ Casimir vacuum + Casimir nuclear
2	Magic Numbers from Substrate Spin-Orbit — $\kappa_{ls} = C_2/n_C = 6/5$	~95% (D-tier, 7/7 magic numbers exact)	Spin-orbit theorems + Mayer-Jensen anchor
3	Nuclear Shell Model (top 30 isotopes) — BST shell-filling	~85% (T2425+ batch)	30-isotope BST verification table

Ch	Title	Existing BST coverage	Key theorems / toys
4	SEMF Coefficients from BST Integers — a_V, a_S, a_C, a_A, δ	~90% (all 5 SEMF coeffs at <2%)	$a_V = 7\alpha m_p/\pi$; $a_S = 8\alpha m_p/\pi$; $a_C = \alpha m_p/\pi^2$; $a_A = m_p/(4 \cdot \dim_R)$; $\delta = (g/4)\alpha m_p$
5	Halo Nuclei — Li-11, Be-11, B-17, dripline structure	~60% (Task #112)	halo-radius BST predictions
6	Superheavy Island Stability — predicted island via BST	~70% (Task #111)	superheavy-Z BST predictions
7	Atomic Structure (electron shells) — orbital sequence ($2l+1$) = 1, 3, 5, 7 IS BST integer sequence	~95% (D-tier, exact match)	Vol 0 operator zoo → atomic orbitals
8	Hyperfine + Lamb Shift — α^{n_C} substrate-coordinate count (T2476)	~70% (Toy 3496 I-tier PATH ARTICULATED; T2476 partial Mode 5 lift)	T2476 substrate-mechanism; Lamb shift 0.4% via Welton-Bethe
9	Atomic Spectroscopy via BST — Rydberg, Klein-Nishina, Compton at $\alpha^{\{\text{rank}=2\}}$	~60% (T2476 multi-observable verification)	T2476 + Penning trap 2030+ falsifier
10	Atomic Clocks + Time Granularity — SP-30 H1 Sr-clock falsifier	~40% (Lyra T2360 prediction)	Koons tick $t_{\text{Planck}} \cdot \alpha^{\{C_2^2\}} \approx 10^{-120}$ s + Sr-clock precision falsifier

Ch	Title	Existing BST coverage	Key theorems / toys
11	Nuclear Decay (substrate-mediated) — Casey neutron decay winding rearrangement framework	~40% (Task #58 + #72 W-32)	$\tau_n = 878.1 \text{ s} + g_A = 4/\pi$ (Fermi + BST inputs)
12	Connections — Vol 2 inputs + Vol 4 BBN outputs — bridge chapter	~50% (cross-volume map)	input from Vol 2 m_p, m_n, α ; output to Vol 4 cosmological abundances

Approximately 60% existing BST coverage maps to specific chapters above. 40% to develop — primarily Ch 5 (halo nuclei detail), Ch 10 (Sr-clock falsifier development), Ch 11 (full neutron decay mechanism).

Cross-volume reference map

Inputs from Vol 0: - D_{IV^5} + five integers - Operator zoo (charge Q + chirality γ^5 + parity P_{op} + Hamiltonian H_{sub}) - Conservation laws (T2473/T2474/T2475)

Inputs from Vol 1: - Bergman Hilbert space $H^2(D_{IV^5})$ - QFT scattering formalism - Renormalization (radiative corrections enter Ch 8 Lamb shift, Ch 9 spectroscopy)

Inputs from Vol 2: - $m_p = 938.272 \text{ MeV}$, $m_n = 939.565 \text{ MeV}$, $m_p/m_e = 6\pi^5$ at 0.002% - $\alpha = 1/137.036$ (fine structure constant) - Particle masses + couplings

Outputs to Vol 4 (GR & Cosmology): - Nuclear binding energies → BBN element abundances - Atomic structure → cosmological recombination - Atomic clocks → cosmological tests of constant constancy

Gap registry (what's missing for full Vol 3 v1.0)

- Ch 5 Halo Nuclei: only Task #112 outline exists; needs full BST treatment of Li-11/Be-11/B-17 structure with substrate mechanism
- Ch 8 Bethe-log substrate-derivation: B6 Lamb shift v0.2 has α^{n_C} I-tier PATH ARTICULATED but full mechanism is multi-week (per Cal #99 + Calibration #21)
- Ch 10 Sr-clock falsifier development: Lyra T2360 prediction exists; needs full experimental design + observable derivation
- Ch 11 Neutron decay mechanism: Casey “winding rearrangement” framework needs full BST derivation (Task #58)
- Ch 12 cross-volume bridge: needs explicit map from Vol 3 outputs to Vol 4 BBN inputs

Status declarations

- v0.1 SCAFFOLD — INDEX + Architectural Scaffold filed, chapter outline drafted, cross-volume map drafted
- v0.3 chapter-grade narrative — when each chapter has chapter-grade content (~10-12 chapters at v0.3 each)

- v0.4 reader-grade pedagogy — 3-level pedagogy applied (Level 1 essence + Level 2 graduate precision + Level 3 5th-grader paragraph)
- v1.0 chapter-grade content REACHED — six-gate framework closed (Content + Cal cold-read + K-audit + Verification toys + Catalog backbone + Calibration #19 sweep)

Saturday 2026-05-23 EOD target: v0.1 SCAFFOLD COMPLETE; team can begin chapter content next.

— Casey Koons + team, Vol 3 Nuclear & Atomic Physics SCAFFOLD v0.1